L-Università ta' Malta
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CPS3235 Data Science: From Data to Knowledge

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Declaration

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CPS3235 — Data Science: from data to knowledge

Course Code — Title of work submitted

25/01/2023

Date

Contents

1 Introduction

All the tasks we run on Lubuntu using the CPS3235 environment provided on VLE. In each task extra libraries were installed to reach the goal of the questions. These are listed in the Readme file.

2 Task 1

2.1

Using the tweepy library all the following and about 33,000 followers from Elon Musk's twitter account were retrieved. Using the json library they were stored in the following text files:

- Followers_raw
- Following_raw

This collection includes the name and username of the respective user. In this design only the username was stored in the database. Using the snscrape library and the twitter module, all Elon Musk's tweets from 01-01-2018 to 10-12-2022 were collected with the following metrics:

- Text
- Likes
- Retweets
- Replies
- Mentions
- Date

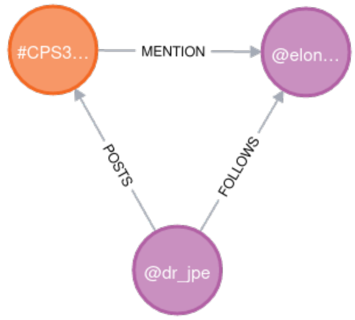

Figure 1: Example to show the design of the graph.

The next step consists in data cleaning. From the following and the followers, the username was selected. Regarding the tweets, the 'Date' column was dropped, and the column 'Mention' was cleaned to have just the username of user mentioned in the same shape of the followers and following.

The same tweets collecting and cleaning process was applied for the dr_jpe's tweets. The final step consists in storing all the data in neo4j. The design includes the following properties:

- A node for users
- A node for tweets
- A relationship called 'FOLLOWS' which links a user to another user if the first follows the second one.
- A relationship called 'MENTION' which links a tweet to a user if the latter is mentioned in that tweet.
- A relationship called 'POSTS' which links a user to a tweet if the latter was written and posted by that user.

Figure 1 shows an example, inspired by task E, of the design of the graph.

2.2

My repository: https://github.com/michelepag/CPS3235-twitter-data.

Ruan's repository: https://github.com/ruanchaves/CPS3235-Twitter-Data-Collection

Mariah's repository: https://github.com/mariah-zm/cps3235-raw-data

All the followers from Ruan's data have been used and 100.000 from Mariah's.

2.3

In this case famous and controversial are quite close to share the same definition. It was considered to select famous tweets looking at the most liked ones and the most retweeted. Regarding controversial, the number of likes was not considered, on the other hand the number of replies and retweets were estimated as valid metrics. However, among the most liked tweets there are also the most controversial.

This is the most liked and retweeted tweet (2). It is also the third one for replies. For sure it is famous and even controversial

This is the second most liked and retweeted tweet (3). It is also the fourth one for replies.

This is the third most retweeted tweet(4).

This is the most replied tweet (5).

This is the second most replied tweet which is very controversial (6).

Figure 2:

Figure 3:

2.4

As it is visible from the world cloud representation 7, Elon Musk's most used words are strictly related to his work and his companies. In fact, we can find among them: Tesla, SpaceX, rocket, Starlink, Mars, future, production, engine, car.

2.5

Query to find a link between @dr_jpe and @elonmusk:

```
MATCH p=(n:User {username:"@dr_jpe"})-[*]->(n1:User {username:"@elonmusk"}) RETURN p
```

Figure (8) shows the result of the query:

2.6

To complete this task using relation databases, I would create three tables. One for Elon's followers, one for the following and one for the tweets. The first two tables will contain the same properties of the node User, and the third one the properties of the node Tweets.. To create the respective relationships, the tables need to have some primary keys. For Following and Followers the username would be enough since it is unique. For the tweets table the id of the creator could be used as a primary key. The relationship mention could be generated by taking the primary of followers and following, and relate it to the mention column of the tweets table, therefore performing a JOIN operation.

The advantages of a relation database are:

Figure 4:

Figure 5:

Figure 6:

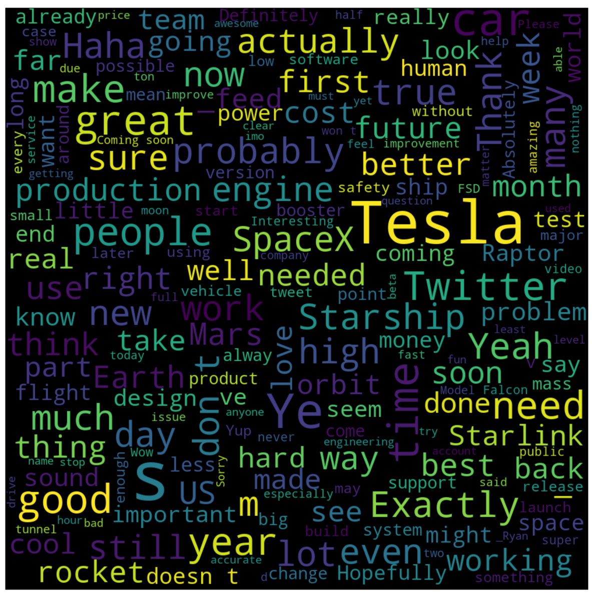

Figure 7: Elon Musk's most used tweeted words

- Generally they offer more data integrity.
- Better handling of duplicates.
- Security.

Disadvantages:

- It is more complicated to create relationship.
- It is not possible to retrieve using paths and shortest paths.
- The operations require a lot of memory storage.
- They have less scalability.

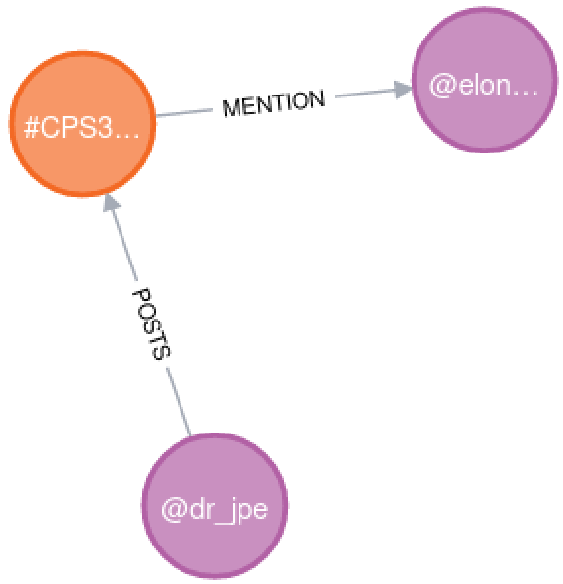

Figure 8: Link between dr_jpe and elonmusk

3 Task 2

Link to the PGN file and csv file: https://drive.google.com/drive/folders/1gr6_XGC8vzlfsNqYHGPWD2VbFXukey-d?usp=sharing

To read the PGN file, the chess-pgn library has been installed.

The collection of the data was not trivial. Looping through the whole file, which includes about 5.6 million games, takes a lot of hours. Several issues were encountered during the loop, such as errors, storage problems and some crashes. Under these circumstances the collection of the data was not run on a Linux environment. The result of the collection is stored in the final_chess.csv file. The latter contains 3.691.281 games and ends with some games from 2010. Hence the timeliness of the dataset is limited.

3.1

One of the interesting aspects to visualize on a heatmap is displaying the squares where is more likely to make a check. The heatmap (9) shows that it is very unlikely to check the opponent's king from the margins of the board. On the other hand, the middle of the board seems to be most dangerous area, reaching its peak in f6.

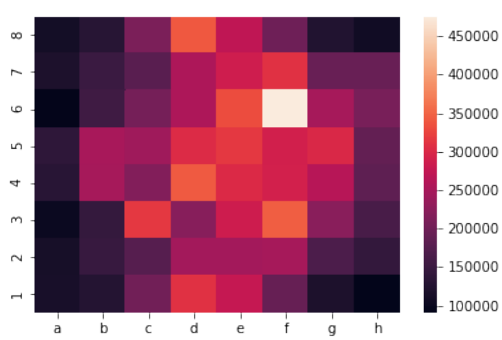

Figure 9: Square from where more checks are made

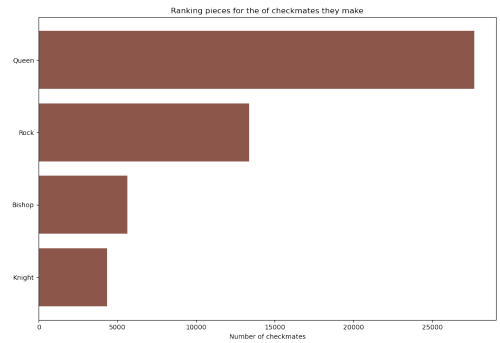

Figure 10: Ranking pieces for the number of checkmates they make

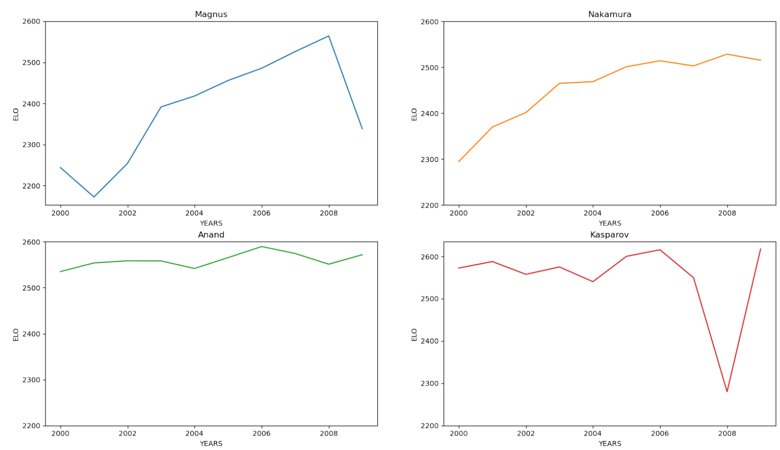

Figure 11: Elo trend from 2000 to 2009

Another captivating question to answer is: What pieces are used the most to reach a checkmate? As shown in figure (10), the Queen is the in top of this ranking. It is not a coincident since the Queen is the most powerful piece. It follows Rock which is more effective than Bishops and Knights. These two have almost the same efficiency, but Bishops prevails on Knights. Regarding King and Pawn, they registered a count of 21 and 0 respectively, hence it was decided to keep them out from the ranking chart.

It was very fascinating to analyse the trend of the ELO of four of the strongest chess players from the 2000s: Magnus Carlsen, Nakamura Hikaru, Garry Kasparov and Viswanathan Anand. The timeliness of this analysis goes from 2000 to 2009.

The four time series plot ((11)) illustrates different things. First of all, in that period the man to defeat was Viswanathan Anand. His graph is extremely constant, which is a synonym of a repetition of great results. In fact he was the world champion from 2007 to 2013 and FIDE champion from 2000 to 2002. Second, we can observe the rises of two of the most skillful players of the current generation: Magnus Carlsen and Nakamura Hikaru. The first one has experienced a huge increase of the elo from 2001 to 2008, while Nakamura's graph shows a more moderate arise. Garry Kasparov is one of the best chess player of all time, reigning champion from 1985 to 2000. This explains the high elo he maintains until it starts to decrease after 2005, year of his retirement.

One of the most intriguing move a chess player can make is called Castling. There are two types of it, kingside and queenside. Figure (12) compares how many times a castling happened in match which eventually ended in a victory. It is clear that the queenside castling is quite less used than the kingside one.

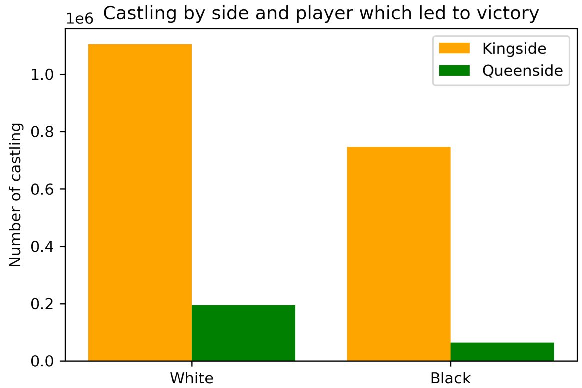

Figure 12: Castling that lead to victory

3.2

For this question a descriptive statistical analysis was performed. Looking at the mode of white and black players, the same name pops up. Hence, most of the games in the data are played by Korchnoi Viktor. The most common result is 1-0. Most of the matches took place in 2003. From the plots shown in figure (13) and (14) it is possible to observe that the mean (red line) and median (green line) are so close to overlap. This means that the variable elo is symmetrically distributed.

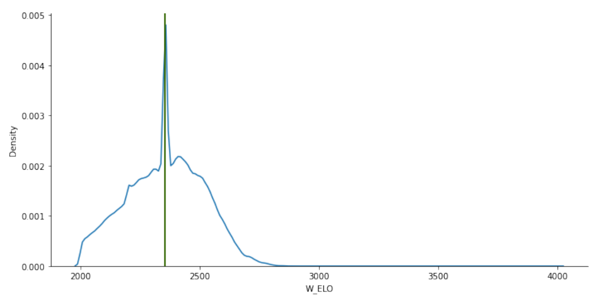

Figure 13: Distribution with mean and median of the elo of white players

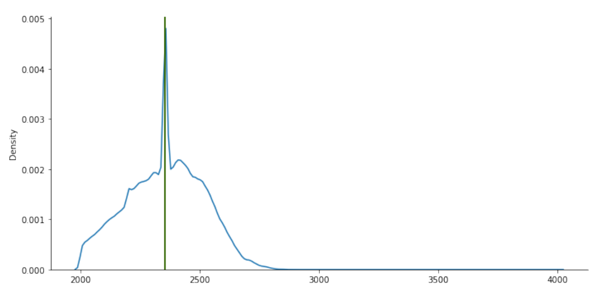

Figure 14: Distribution with mean and median of the elo of black players

The variance of the elo of white players is about 25.322 and for black ones 21.612, while the MSE is 159 for white and 147 for black. These results warn us that there are values which differs a lot from the mean, hence possible outliers. This is visible also from the plots.

3.3

The image shown in figure (15) is a visual representation of the results of a survey made by EMG. This organisation asked about 1500 Italian people if they agreed to send weapons to Ukraine to support the country for the recent war outbreak. First of all a pie chart was chosen to represent the data. Usually it is better to avoid pie charts and use bar charts instead, because it is difficult to read individual values. Given that, the evident mistake lies on the proportions of the graph. The green area is visually smaller than the sum of the other two, even though the green area should represent 55% of the total area.

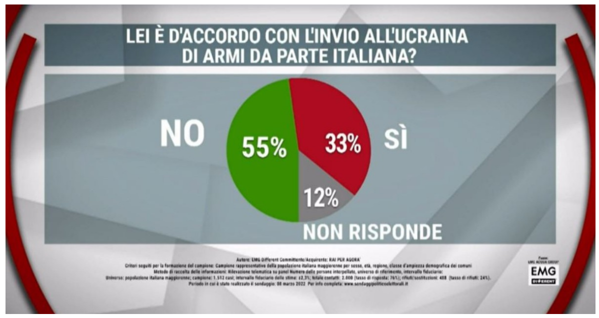

Figure 15:

The pie chart was replaced by a more appropriate bars chart (16).

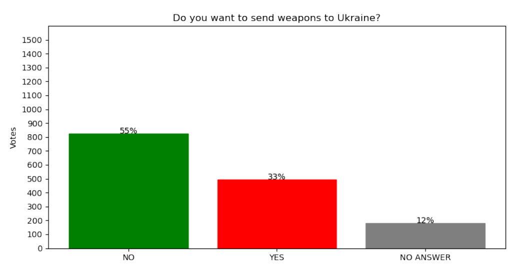

Figure 16:

4 Task 3

4.1

The data provided has a single source which is the Times of Malta. The domain can be identified as Properties for sale. The data lists properties from 2015 to 2022. There are 323 files with more the 100.000 properties listed. The collection was quite easy since the ul html elements where the properties are listed are identified by the classes named 'classified_list' and 'ca_Classifieds_PostList'. The data has multiple mistakes and there several inconsistent elements. For instance, the data contains some properties from Sicily or announcement of price reductions. Not all the properties are listed with a price, a sqm vale , the number of bedrooms, the type of property. Hence the final dataset featured a lot of missing values. Some properties are listed with a price in million instead of thousands or sometimes with not realistic prices. Sometimes the number of bedrooms is written in words and not in numbers. Although the high presence of mistakes and inconsistencies, the data contains a lot of information.

4.2

The selected features for the dataset are:

- Price. Continuous variable.

```python
#check missing values
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 138915 entries, 0 to 138914
Data columns (total 8 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   LOCATION          135341 non-null  object
 1   PRICE             119233 non-null  float64
 2   TYPE_OF_PROPERTY  123235 non-null  object
 3   POOL              138915 non-null  object
 4   GARAGE            138915 non-null  object
 5   DATE              138915 non-null  object
 6   BEDROOM           69056 non-null   object
 7   SQM               27293 non-null   float64
dtypes: float64(2), object(6)
memory usage: 8.5+ MB
```

Figure 17: Number of not missing values

- SQM (Square per meters). Continuous variable.
- Number of bedrooms. Categorical qualitative variable.
- Type of house. Categorical variable.
- Date. Categorical variable.
- Garage. Binary variable.
- Pool. Binary variable.

First, the goal was to extract the more features as possible. It was opted to get at least two features per type and to have some features with a probable correlation.

Looking at the number of missing values (17, it is clear that price, location and type of property have a small percentage of missing values. Therefore, it was opted to remove all the rows with missing values. This procedure ensures the removal of inconsistent data since it is unlikely to find a valid announcement without price or location.

On the contrary, bedroom and sqm have a high percentage of missing values. Even though is good practice to remove columns in these scenarios, all the missing number of bedrooms values have been replaced with the most common value (i.e., 3). Since the percentage of missing values for SQM was too high, it was decided to leave the column like that. It was not dropped because it could have been useful for some statistical analysis.

Price had different outliers, most of them unrealistic values. Hence, when it came to clean the data, all the prices under 10.000 were considered missing values. This decision was built also on the fact that we are handling properties which can unlikely be estimate with such a low value. Location was the most troublesome feature to clean given the high presence of outliers and mistakes in the data. Therefore, only the top 80 location with more properties on sales were selected. The appropriateness of this choice lies on the small reduction of entries the dataframe faced, going from 107.549 to 103.853.

4.3

What are the locations with more properties on sale? Figure 18 shows the top 15 locations for properties on sale.

Apparently Sliema and Gozo have about 2000 more properties than Mosta and three times the properties on sale in Zejtun.

What are the locations with more properties with pools on sale?

Apparently almost a third of the properties on sale in Gozo have a pool. Let's see the data without Gozo.

Figure 20 shows the same scenario, but without Gozo.

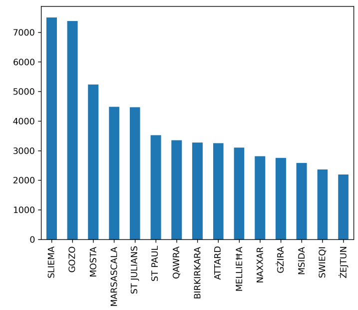

Figure 18: Location with the most number of properties on sale

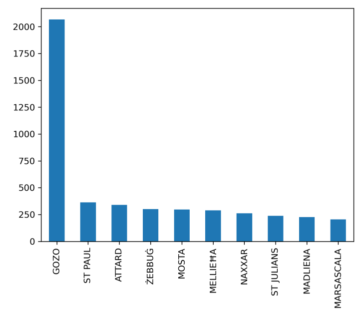

Figure 19: Location with the most number of properties on sale with a pool

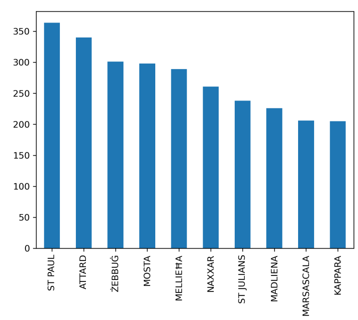

Figure 20: Location with the most number of properties on sale with a pool

In the main island, St Paul has more properties with a pool on sale. It's interesting to see that Sliema, which is the top location for number of properties on sale, in this context it is not even in the first ten.

How have the prices changed from 2015 to 2022?

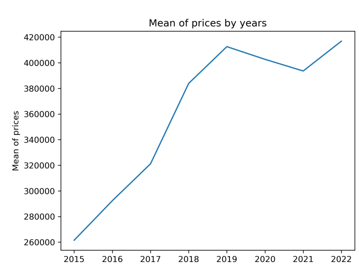

Figure 21: Change of the mean of price during the years

This graph shows how in the last 8 years in Malta the price of the properties has increased drastically. The only moment where prices decreased a bit is between 2019 and 2021. This variation was probably caused by the COVID-19 pandemic.

What type of property is more likely to have a garage and a pool?

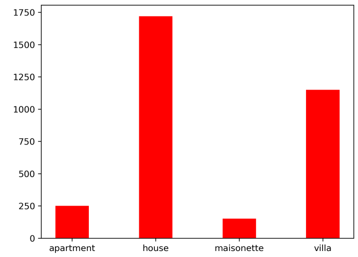

Figure 22:

Villas and Houses are more likely to more likely to have a garage and a pool compared to maisonettes and apartments.

4.4

A correlation has been found between the binary feature POOL and the continuous SQM. To find a correlation between these two kinds of features, a point biserial test was conducted. (23).

```
stats.pointbiserialr(no_miss_sqm['POOL'], no_miss_sqm['SQM'])
PointbiserialrResult(correlation=0.2566085938529463, pvalue=2.375490331610222e-288)
```

Figure 23:

Since the p-value is less than 0.05 there is a significant statistical correlation between the two variables. The statistical variable suggests a weak positive correlation, hence more sqm a properties has, the more likely is to have a pool.

4.5

It was decided to perform a statistical inference analysis. Specifically, a T-test was performed with the following hypothesises: H0: There is no difference in the mean of the price. H1: There is a difference in the mean of the price. The two samples are the following:

- 30 randomly picked houses with no garage, no pool, in ST JULIANS and from 2015.
- 30 randomly picked houses with no garage, no pool, in ST JULIANS and from 2022.

The t-test needs to respect some assumptions.

The variable is normally distributed in both samples. To confirm this assumption a Shapiro test is performed for both the samples (24).

```
from scipy.stats import shapiro
print(shapiro(new_df_2015['PRICE']))
print(shapiro(new_df_2022['PRICE']))
ShapiroResult(statistic=0.7764552235603333, pvalue=2.5123990781139582e-05)
ShapiroResult(statistic=0.7611837387084961, pvalue=1.3879584912501741e-05)
```

Figure 24: Result of shapiro test

The p-values for both samples are less then 0.05. It is safe to conclude that the variable is normally distributed in both samples.

The two samples have similar variance 25

All the assumptions are satisfied, and the t-test can be performed.

As figure 26 shows, since the p-values is less than 0.05, the null hypothesis is rejected. Hence there is a significant difference between the means of price of the two samples.

```python
var1 = new_df_2015['PRICE'].var()
print(var1)
var2 = new_df_2022['PRICE'].var()
print(var2)
```

```
33099774712.64369
42796695402.29886
```

Figure 25: Variance of price in the two samples

```python
stats.ttest_ind(a=new_df_2022['PRICE'],b=new_df_2015['PRICE'], equal_var = True)
```

```
Ttest_indResult(statistic=2.192932939147687, pvalue=0.03233627550790334)
```

Figure 26: Result of t-test

4.6

The predictive model was created thinking at real world problem scenario. Consider a person who owns an apartment in Sliema and they are considering to put it on sale. They would like to get an estimate of the prices looking at the other properties which share the same features. Hence, it was decided to predict the price of an apartment in Sliema with no garage and no pool in 2022.

Looking at the correlations, the only significant one is between Price and SQM, as shown in the scatter plot 27. To confirm this correlation a person correlation test has been performed. The p-value is 1.2195842632203556e-11 which means that there a significant correlation between the two feature since it is less than 0.05. The statistical value is 0.7822160576279074, therefore the relationship is positive and strong since the value is close to 1.

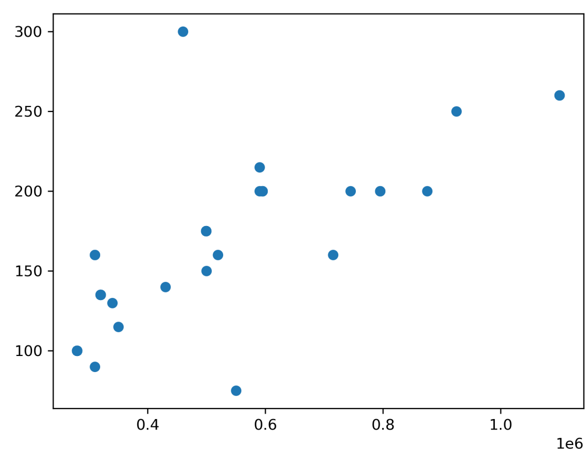

Figure 27: Scatter plot between price and SQM

The dataframe was reduced to 50 elements given the conditions proposed. The variable to predict is Price and SQM is the only predictor. The model chosen to predict the variable is a Simple Linear Regression

Using the sklearn library, the data was split in train and test sets with a 70/30 percentage. Once the model was fit and the predictions computed, the accuracy of the predictive model was computed using as parameter the MSE. This value is extremely high (45212910762.33178), therefore the model is quite inaccurate.

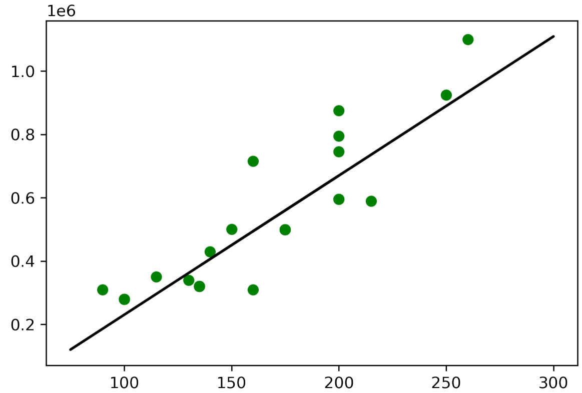

Figure 28: Linear regression between price and SQM

5 References

- https://stackoverflow.com/questions/45758646/pandas-convert-string-into-list-of-strings
- https://stackoverflow.com/questions/66053317/how-can-i-plot-a-heatmap-from-my-python-dictionary
- https://stackoverflow.com/questions/1454913/regular-expression-to-find-a-string-included-betwee
- https://www.geeksforgeeks.org/python-convert-numeric-words-to-numbers/
- https://betterprogramming.pub/detecting-sentiment-from-elon-musks-tweets-using-python-ec7820469